

AKARSH KUMAR

Full Stack Engineer

akarshdbg@gmail.com | +91 9693061651 | Noida, India

[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

SUMMARY

Full Stack Engineer with hands-on experience building production-ready web applications using the MERN stack, along with exposure to microservices architecture and containerization with Docker. Skilled in designing scalable REST APIs, real-time WebSocket systems, and secure authentication flows. Comfortable working across the stack, from React frontends to backend services.

PROJECTS

Codox – Real-Time Collaborative Code Editor

Sep 2025 – Oct 2025

React, Node.js, Express, Socket.IO, Redis, Google OAuth [GitHub](#) | [Live Demo](#)

- Engineered an end-to-end real-time collaboration platform using a Socket.IO backend and a React frontend consuming WebSocket events
- Implemented OAuth 2.0 authentication with Redis-backed session storage across backend and frontend routing
- Architected a room-based system supporting multiple concurrent editing sessions with sub-50ms backend-to-client sync

API Gateway with Express & Docker – Microservices Project

Jan 2026

Express.js, FastAPI, Docker, Docker Compose, Postman [GitHub](#) | [Postman Docs](#)

- Designed a microservices architecture with a centralized Express.js API Gateway routing requests to independent Blog (FastAPI) and User (Express.js) services
- Containerized 3 services (API Gateway, Blog, User) using Docker and Docker Compose, enabling isolated inter-service communication over a custom Docker network
- Documented and tested all REST API endpoints across both microservices using Postman, publishing public API documentation for the gateway

Satvik AI – AI-Powered Nutrition Platform

Jul 2025 – Aug 2025

React, Node.js, Express, MongoDB, Redis, Tailwind CSS, Gemini API [GitHub](#) | [Live Demo](#)

- Developed a full-stack MERN application with a RESTful API backend serving role-based endpoints to a React + Redux frontend
- Reduced API response times by 60% by implementing Redis caching, improving frontend data loading
- Integrated an asynchronous Gemini AI pipeline to process user uploads and return structured nutrition data

Tasko – Task Management Application

Mar 2025 – Apr 2025

React, Node.js, Express, MongoDB, Redis, Tailwind CSS, Gemini API [GitHub](#) | [Live Demo](#)

- Built a full-stack system with JWT authentication on the backend and protected React routes consuming secure endpoints
- Improved frontend load times from 200ms to 15ms by optimizing data flow with Redis caching
- Delivered an AI-powered subtask generation feature integrating the Gemini API with a dynamic React UI

SKILLS

Languages:	JavaScript, Python, C++
Full Stack:	MERN Stack (MongoDB, Express.js, React.js, Node.js), REST APIs
Backend:	Node.js, Express.js, FastAPI, WebSocket/Socket.IO, Event-Driven Architecture, API Integration
Frontend:	React.js, Redux Toolkit, React Router, Tailwind CSS, State Management
Infrastructure:	Docker, Docker Compose, Microservices Architecture, API Gateway, Postman
Auth & Security:	JWT, OAuth 2.0, Session Management, Token Rotation, Access Control
Databases:	MongoDB, Redis, Query Optimization, Performance Tuning
Tools:	Git, GitHub, VS Code, Data Structures & Algorithms

EDUCATION

Techno India University, Kolkata

2022 – 2026

B.Tech in Computer Science & Engineering (AI)

CERTIFICATIONS

- Python (Basic) – HackerRank
- SQL (Basic) – HackerRank